Assessment for Instruction: Noyce Scholar Use of Student Performance Data

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Introduction and Methods

Since 2015, UVM's Noyce program has awarded 25 Noyce Scholarships to MAT students who are committed to teaching math and science in high need schools. The program functions as a cohort model whereby students take required classes together beginning in mid-July and continuing through May of the following year including a year-long school-based internship. Students apply to both the MAT and Noyce programs and are awarded a scholarship based on academic merit, need, and professional experience. As a requirement for the VT Licensure Portfolio, MAT students engage in an **Analysis of Learning Project** during the 4-week solo period requiring them to reflect upon how their teaching influences student learning and growth over time. This study focuses on the experiences of a Noyce scholar enrolled in the MAT program at the University of Vermont. Case study research methodology (Stake, 2006) was utilized to address the question: ***How effective are Noyce scholars in utilizing student performance data to adapt and modify instruction?***

An interpretivist case study approach (Stake, 2006) was utilized to seek meaning and understanding of scholar experiences in context. Multiple sources of data such as interviews, observations, and artifacts were collected for the construction of vignettes that illustrate facets of the case and represent new findings and understandings for each of the participants. For this study, we selected one case for analysis because it was among the most detailed and representative of the best practices embedded in the MAT program of study.

Case Study Analysis

A Case Study of Meredith

"Meredith" (pseudonym) was a biology major who completed the one-year MAT internship in a local public HS. She earned a 4.0 GPA in the MAT Program and 189/200 in the General Science and 187/200 in the Biology Praxis II Exams. She secured a science teaching license (7-12), a science teaching position in a high need school in VT and completed the 2-year NSF teaching commitment.

As part of state licensure portfolio requirements, students complete an **Analysis of Learning Project** during the 4-week solo teaching period that demonstrates their ability to summarize evidence for student growth aligned to learner outcomes and to reflect upon how their teaching practice may have influenced student learning over this time period.

For her analysis project, Meredith describes the Cell Cycles unit that she developed for a 10th grade biology class (20 students) in the following excerpt.

> ***Many of the students have a lot of energy and don't appreciate sitting still. Four students in this class are on IEPs with varying degrees of modifications and accommodations and two students are on 504 plans. This unit includes lessons on mitosis, meiosis and cell differentiation. Students had no background knowledge of these processes, but they knew from our study of cell theory that all cells arise from preexisting cells and had some basic knowledge about DNA from middle school. The unit touches upon how and why new cells are made and how cells appear differently to become different components of an organism. Lastly, with meiosis, students learn a major source of genetic diversity and how organisms make cells necessary for sexual reproduction.***

The goal of the pre/post assessment attempted to assess student understanding of mitosis, meiosis, and cell differentiation. In asking the question: "How do humans become 100 trillion cells, yet start as one cell?" students needed to recognize that meiosis is the process by which gametes are formed resulting in half the number of chromosomes of typical somatic cells and that mitosis (not meiosis) is the process that explains how humans start as 2 gametes (with half the number of chromosomes) to become an organism comprised of 100 trillion cells.

Analysis of Learning Project

Tracing student progress/development using assessment data is a core responsibility of every teacher. At any given point in time, a teacher must be able to answer the question: How do I know what my students know? The Analysis of Learning demonstrates your ability to summarize evidence for student growth along major learner outcomes in your unit plan and to reflect upon how your teaching practices have fostered (or failed to foster) student learning and growth. This is a required piece of evidence for Part II of your Vermont Licensure Portfolio.

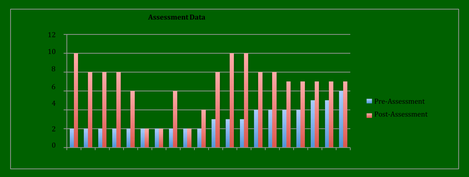
The analysis of learning will be structured around the unit plan that you include in Part II of your portfolio. If you teach this unit to more than one section of students, you will only need to analyze assessment data for one class.

The analysis of learning is organized into five major sections:

Part I: Overview of Unit/Learner Outcomes
Part II: Assessment Strategies
Part III: Learner Profile
Part IV: Data Display and Analysis of Data
Part V: Reflective Narrative

(Developed by Barri Tinlker, 2018)

Table 1- Pre/Post Assessment Data



Reference:
Stake, Robert E. (2006). *Multiple case study analysis*. New York, NY: Guilford.

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https://www.uvm.edu/~noyce/

Case Study Analysis

Table 1 illustrates pre and post assessment results from 20 students in this 10th grade biology class. In the following excerpt, Meredith reflects on these results.

> ***Based on my data of the pre and post assessment (Table 1), sixteen students met the learning objective of identifying and explaining the role of chromosomes and modeling mitosis. Only 3/20 students correctly connected cell differentiation to this prompt, meeting the learning objective of modeling and explaining the role of cell differentiation. Another six students were close to making this connection, but they either included meiosis in their response or spoke about cell differentiation as a way to provide genetic diversity. This tells me that these students were unable to correctly identify the difference and similarities between meiosis and mitosis. I can't know from this data if the other students who didn't mention meiosis were able to meet this learning objective. The absence of meiosis in their responses may mean they know to distinguish meiosis as an unrelated process, or it might imply they aren't comfortable with the process of meiosis.***

Meredith reflects on her decision about the unit test format and her rationale for utilizing a variety of assessments in assessing student knowledge of unit concepts in the following:

> ***Though I would have liked to just give constructive response questions, I wanted to give students who struggle with writing an opportunity to show their knowledge and understanding in this unit. Some students have a lot of knowledge but are challenged in expressing this in an open-ended way. My post-assessment matched to the pre-assessment was an open-ended constructive response assignment that works well for some students, but is challenging for others. Therefore, I examine that data, but I also keep in mind the data from other formative and summative assessments included in this unit.***

Summary and Conclusions

Given the opportunity to teach this unit again, Meredith shared that she would place more emphasis on differentiating her teaching and assessment practices and that her strengths were largely focused on her ability to differentiate by student readiness and to a lesser extent on content and student interest. In the end, she still felt that the strategies that she employed approached learners in the same way.

> ***Many of the students who show the most learning are students that thrive in a traditional school environment. I believe this was because even with intentions of creating meaningful, engaging and hands-on lessons, I tended towards teaching content in the traditional manner. I made an effort to incorporate different styles of teaching and allow students to work through content in a variety of ways, but my default form of imparting content knowledge is still direct instruction. Some students struggled to meet the learning objectives because I didn't give them enough of an option to learn the content in diverse ways.***

> ***Though my intentions are to address the needs of all learners, I see in my data that this isn't always the case. Some redesign of lessons to use more inquiry based and constructive approaches would help to hook students and lead to more powerful learning. I need to remember not just that students learn differently, but that any subject can be taught in a number of ways. By exploring my own creativity and diversifying instructional approaches, I can maximize student learning and create lasting lessons for students.***

In summary, Meredith used both formative and summative assessments at the individual level as well as assessments administered to the whole class to gauge continuous progress and identify opportunities to redirect and/or modify the instruction. She utilized quantitative and qualitative analytical methods to describe the results of instruction while accounting for such variables as IEP status and students' prior knowledge. The analysis and presentation of data by Meredith is quite sophisticated and clearly documents learning that individual students, groups of students, and the whole class demonstrated over a 4-week period. An important dimension of this case is the analysis of instruction that she presented following the post test. This case clearly demonstrates that Meredith was able to utilize student performance data to inform the way that she would adapt and modify instruction over time.